

Practical 0

* Aim :- Introduction to PHP Lab.

* Theory :- PHP (Hypertext Preprocessor) is a popular server-side scripting language used for developing dynamic and interactive web applications. It was originally created by Rasmus Lerdorf in 1994. PHP is designed especially for web development, and its code can be embedded directly into HTML. Unlike client-side languages, PHP programs are executed on the web server, and the resulting output is sent to the user's browser in the form of HTML.

PHP is an open-source language, which means it is freely available and can be modified or distributed by anyone. Because of this, it has a large community of developers who continuously improve the language and provide support. PHP is widely used for creating websites, web applications, login systems, from handling session management and database-driven applications.

One of the major advantages of PHP is its ability to interact with databases. It works efficiently with database management systems such as MySQL, allowing developers to store, retrieve and manipulate data easily. This feature makes PHP very useful for

building dynamic websites where information is frequently updated.

PHP is platform-independent and can run on various operating systems such as windows, linux and macOS. It is compatible with most web servers, including Apache HTTP Server, and can also work with servers like Nginx. Because of its flexibility and ease of use, PHP is widely adopted in the web development industry.

PHP is also used in many well-known content management systems and web platforms such as WordPress, Drupal and Joomla. Because of its powerful capabilities and simplicity, PHP remains one of the most widely used languages for web development today.

* Conclusion :- From this practical, we learned the basic concept of PHP and how it is used as a server-side scripting language to develop dynamic web pages. We also understood how PHP can be integrated with HTML and databases to create interactive web applications. PHP is easy to learn, flexible, and widely used in modern web development.

Practical 1

Aim:- Create a basic PHP script that prints "Hello World".

* Theory:- PHP is a widely used server-side scripting language designed for web development. It allows developers to create dynamic and interactive web pages. PHP code is executed on the server, and the output generated by the script is sent to the user's web browser in the form of HTML.

A basic PHP program is written inside special PHP tags `<?php` and `?>`. These tags tell the server that the code inside them should be processed as PHP code. PHP provides various built-in functions that make it easy to perform tasks such as displaying output, processing form data, and interacting with databases.

One of the most commonly used commands in PHP is `echo`, which is used to display text or data on the web page. In a basic program, the `echo` statement can be used to print a simple message like "Hello World" to the browser. This program is often the first example used when learning a new programming language because it demonstrates the basic syntax and execution of the language.

PHP works efficiently with web servers such as Apache HTTP server and databases like

MySQL. These technologies are commonly bundled together in packages like XAMPP, which provides a local server environment for testing PHP programs.

Creating a simple "Hello World" program helps beginners understand the structure of a PHP script, how PHP code is embedded in a file, and how the server processes the code to generate output.

* Conclusion :- In this practical, we created a basic PHP script that prints "Hello World" on the web browser. Through this exercise, we learned the basic structure of a PHP program, the use of PHP tags, and the echo statement for displaying output. This practical provides a fundamental understanding of how PHP scripts work and servers as the first step toward learning more advanced PHP programming concepts.

Practical 2

* Aim:- Write a PHP script demonstrating the use of variables, data types, and constants.

* Theory:- PHP is a server-side scripting language used to develop dynamic web pages. It allows programmers to store and manipulate data using variables, data types and constants.

In PHP, a variable is used to store data that can change during the execution of a program. Variables start with the \$ symbol followed by the variable name. For example, \$name = "Rishabh"; stores a string value in the variable \$name. PHP automatically identifies the data type of the variable based on the value assigned to it. Common data type in PHP include integer, float (decimal numbers), String (text), and boolean (true or false). Because PHP automatically determines the data type, it is known as a loosely typed language.

Another important concept in PHP is constants. A constant is a value that cannot be changed once it is defined. Constants are created using the define() function. For example, define("SITE_NAME", "My Website"); creates a constant named SITE_NAME. Unlike variables, constants do not use the \$ symbol and their values remain fixed throughout the program.

In PHP, `var_dump()` function is used to display detailed information about a variable. It shows the data type and the value stored in the variable.

Ex:- `var_dump($name);`

- * Conclusion :- In this practical, we wrote PHP scripts to demonstrate the use of variables, data types, and constants. We learned how variables store different types of data and how constants hold fixed values throughout the program. This practical helped to understand the basic concepts of data handling in PHP.